

5 (**long**) A ramjet is an air-breathing propulsion engine that contains no moving parts, as shown in Fig. 4. Air at thermodynamic state 1 enters at speed v_1 , decelerates to thermodynamic state 2, which has negligible speed, is heated at uniform pressure to thermodynamic state 3, which also has negligible speed, and then accelerates to thermodynamic state 4 at speed v_4 . You may assume that air is a perfect gas with $\gamma = 7/5$, $c_p = 1005 \text{ J kg}^{-1} \text{ K}^{-1}$, and $R = 287.1 \text{ J kg}^{-1} \text{ K}^{-1}$.

(a) Show that

$$\frac{T_2}{T_1} = 1 + \frac{v_1^2}{2c_p T_1} \quad [2]$$

(b) Assuming for now that the flow decelerates from 1 to 2 reversibly and adiabatically, derive an expression for the isentropic compression ratio $p_{2_{isen}}/p_1$ in terms of $v_1/\sqrt{2c_p T_1}$ and sketch $p_{2_{isen}}/p_1$ as a function of $v_1/\sqrt{2c_p T_1}$. [6]

(c) Calculate T_2 and $p_{2_{isen}}$ if $T_1 = 200 \text{ K}$, $p_1 = 0.03 \text{ MPa}$, and $v_1/\sqrt{2c_p T_1} = 2$. Comment on your results. What value of v_1 does this correspond to? [4]

(d) In reality, the flow between 1 and 2 decelerates irreversibly and adiabatically and the real compression ratio p_2/p_1 is 20% of the isentropic compression ratio calculated above. Show that the specific entropy increase is $s_2 - s_1 = 462.1 \text{ J kg}^{-1} \text{ K}^{-1}$ and sketch this process on a $T - s$ diagram. [6]

(e) The heat addition per unit mass of air is $q = 1.005 \times 10^6 \text{ J kg}^{-1}$. Calculate T_3 . [4]

(f) If the exit area is carefully chosen such that $p_4 = p_1$ and if the acceleration from state 3 to state 4 is isentropic, sketch the full cycle on a new $T - s$ diagram. [4]

(g) Calculate v_4 and the engine's net thrust per unit mass of air passing through the engine. You may assume that the mass of fuel is negligible. [4]

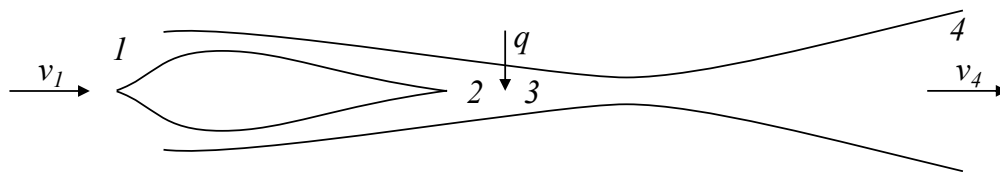


Fig. 4